

Data integration

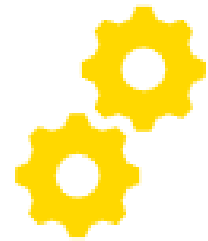
DATA TRANSFORMATION IN KNIME



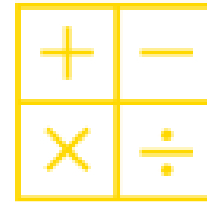
Mahantesh Pattadkal

Sr. Data Scientist

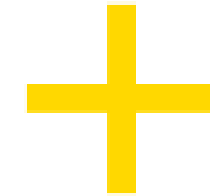
Data integration



Conditional
Statements

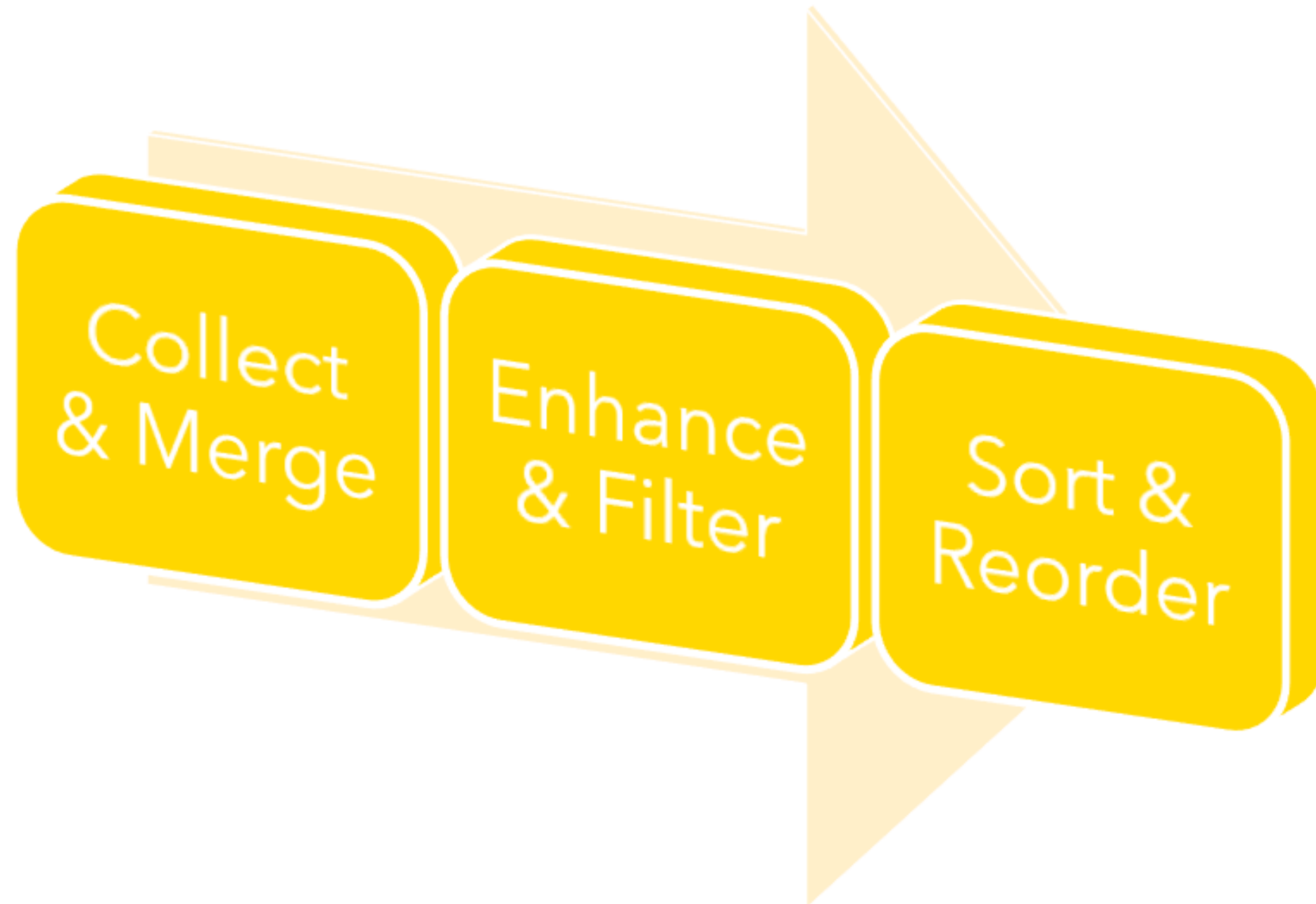


Expression
based
Transformation



Data
Integration

What is data integration?



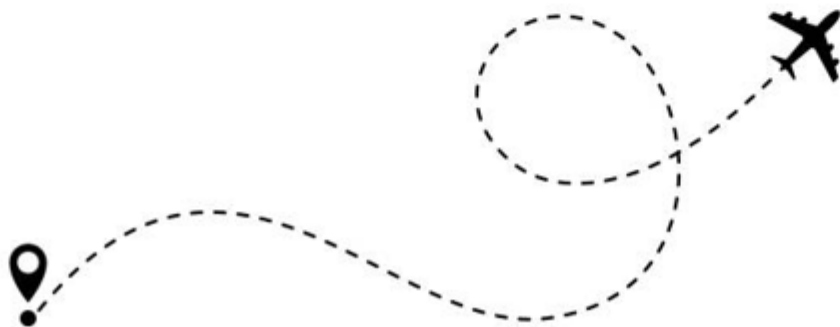
Use case - Air travel

Flight Info

ID	Departure	Price
1A	Berlin	\$1000
1B	Paris	\$500
2A	Berlin	\$1500
2C	Paris	\$2000

Meal Info

ID	Meal1	Meal2
1A	Sandwich	?
1B	Juice	Salad
2A	?	Salad
2C	Fruit Bowl	?



Append columns

Combined Info

ID	Departure	Price	Meal1	Meal2
1A	Berlin	\$1000	Sandwich	?
1B	Paris	\$500	Juice	Salad
2A	Berlin	\$1500	?	Salad
2C	Paris	\$2000	Fruit Bowl	?

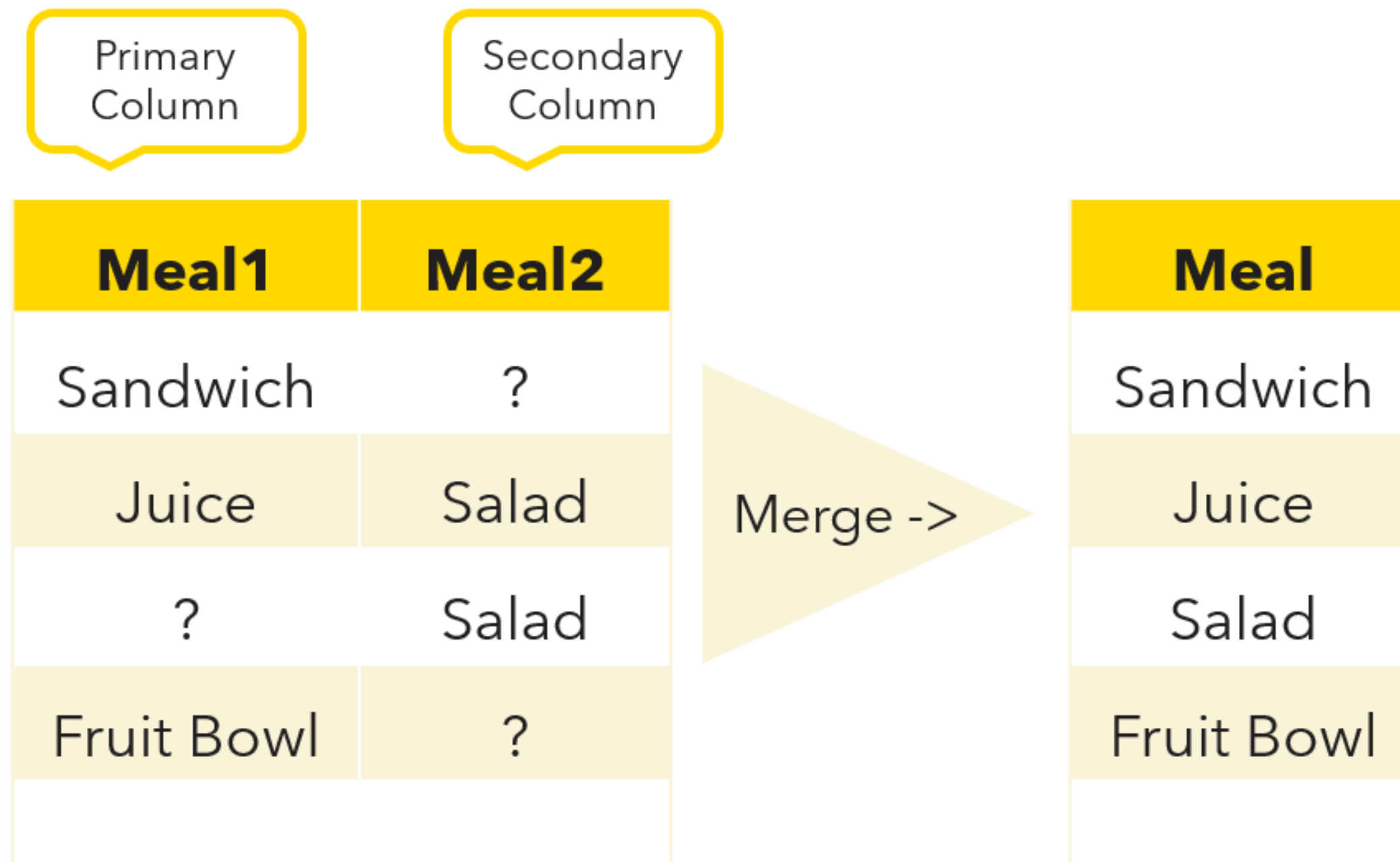
Meal preference

Meal1	Meal2
Sandwich	?
Juice	Salad
?	Salad
Fruit Bowl	?

Meal preference

Primary Column	Secondary Column
Meal1	Meal2
Sandwich	?
Juice	Salad
?	Salad
Fruit Bowl	?

Merge columns



Context

ID	Departure	Price	Meal
1A	Berlin	\$1000	Sandwich
1B	Paris	\$500	Juice
2A	Berlin	\$1500	Salad
2C	Paris	\$2000	Fruit Bowl

All flight
on 7th May

Constant value column

ID	Departure	Price	Meal	Date
1A	Berlin	\$1000	Sandwich	7 May
1B	Paris	\$500	Juice	7 May
2A	Berlin	\$1500	Salad	7 May
2C	Paris	\$2000	Fruit Bowl	7 May

Constant
Value

Constant value column

ID	Departure	Price	Meal	Date
1A	Berlin	\$1000	Sandwich	7 May
1B	Paris	\$500	Juice	7 May
2A	Berlin	\$1500	Salad	7 May
2C	Paris	\$2000	Fruit Bowl	7 May

Reorder column sequence

				Move to first position
ID	Departure	Price	Meal	Date
1A	Berlin	\$1000	Sandwich	7 May
1B	Paris	\$500	Juice	7 May
2A	Berlin	\$1500	Salad	7 May
2C	Paris	\$2000	Fruit Bowl	7 May

Reorder column sequence

Date	ID	Departure	Price	Meal
7 May	1A	Berlin	\$1000	Sandwich
7 May	1B	Paris	\$500	Juice
7 May	2A	Berlin	\$1500	Salad
7 May	2C	Paris	\$2000	Fruit Bowl

Reorder column sequence



Date	ID	Departure	Price	Meal
7 May	1A	Berlin	\$1000	Sandwich
7 May	1B	Paris	\$500	Juice
7 May	2A	Berlin	\$1500	Salad
7 May	2C	Paris	\$2000	Fruit Bowl

Reorder column sequence

Date	ID	Departure	Meal	Price
7 May	1A	Berlin	Sandwich	\$1000
7 May	1B	Paris	Juice	\$500
7 May	2A	Berlin	Salad	\$1500
7 May	2C	Paris	Fruit Bowl	\$2000

Sorting

First Criteria -Sort
alphabetically

Date	ID	Departure	Meal	Price
7 May	1A	Berlin	Sandwich	\$1000
7 May	1B	Paris	Juice	\$500
7 May	2A	Berlin	Salad	\$1500
7 May	2C	Paris	Fruit Bowl	\$2000

Sorting

Sorted by Departure

Date	ID	Departure	Meal	Price
7 May	1A	Berlin	Sandwich	\$1000
7 May	2A	Berlin	Salad	\$1500
7 May	1B	Paris	Juice	\$500
7 May	2C	Paris	Fruit Bowl	\$2000

Sorting - add criteria

Second Criteria -Sort
Descending

Date	ID	Departure	Meal	Price
7 May	1A	Berlin	Sandwich	\$1000
7 May	2A	Berlin	Salad	\$1500
7 May	1B	Paris	Juice	\$500
7 May	2C	Paris	Fruit Bowl	\$2000

Sorting - add criteria

Expensive flight -2A

Date	ID	Departure	Meal	Price
7 May	2A	Berlin	Salad	\$1500
7 May	1A	Berlin	Sandwich	\$1000
7 May	2C	Paris	Fruit Bowl	\$2000
7 May	1B	Paris	Juice	\$500

Let's practice!

DATA TRANSFORMATION IN KNIME

Integrate and analyze students data

DATA TRANSFORMATION IN KNIME



Mahantesh Pattadkal
Sr. Data Scientist

Let's practice!

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